Near-Real-Time Monitoring of Gridded Livestock Methane Emissions Using Multi-Source Data Integration

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Introduction

Methane (CH_4) is a short-lived but highly potent greenhouse gas, possessing a global warming potential over 80 times greater than carbon dioxide (CO_2) over a 20-year timescale. Anthropogenic activities like livestock agriculture constitute the dominant sources of CH_4 emissions. Despite its critical role in accelerating climate change, current CH_4 monitoring systems suffer from major limitations:

- Inadequate spatial and temporal resolution
- Significant latency in data availability
- Over reliance on either bottom-up inventories or top-down atmospheric measurements in isolation

These limitations hinder timely and accurate decision-making for emissions mitigation. There is a pressing need for high-resolution, dynamic monitoring systems that reconcile discrepancies between data sources and facilitate climate accountability.

Objective

To develop high-resolution, gridded maps of anthropogenic methane emissions that bridge the divide between satellite-derived (top-down) estimates and inventory-based (bottom-up) data. By integrating multi-source datasets through machine learning, this project aims to:

- Provide near-real-time detection of high-emission zones.
- Enable more responsive methane policy interventions.
- Support scalable climate accountability and mitigation frameworks.

Methods and Materials

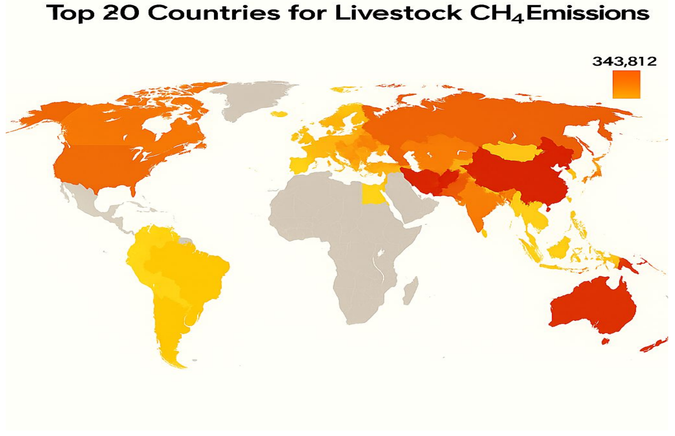
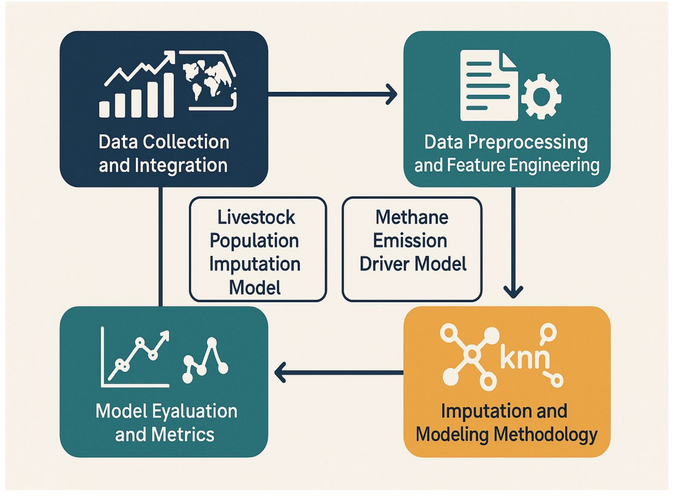


Figure 1: *Top 20 highest-emitting countries visualized using GeoPandas, based on cleaned data from EDGAR, CEDS, and FAOSTAT, to identify global emission patterns and regional trends.*



Model Evaluation and Credibility Assessment

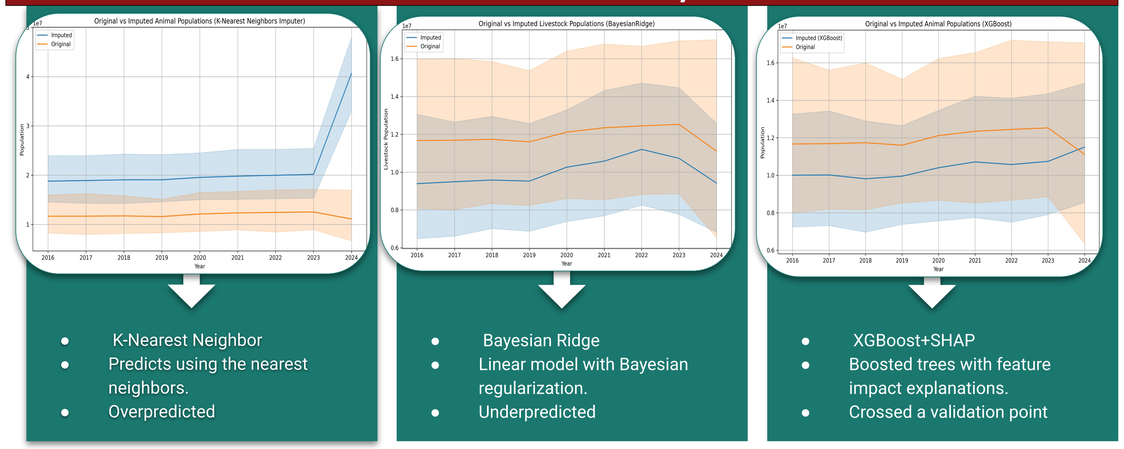


Figure 2: *Comparison of imputed animal populations using KNN, Bayesian Ridge and XGBoost methods, visualizing original vs. imputed data over time to assess the performance and credibility of each imputation technique.*

Model Selection and Performance Evaluation

```
Training rows: 2741
Rows to predict: 887
✅ Validation RMSE: 22,791,185
✅ Validation R²: 0.7387
✅ Final imputed dataset saved as Deploying_New_Livestock_Count_Imputed.csv
Sample of imputed rows:
    Livestock_Count  Imputed_Flag
46     1.669917e+06          True
47     2.776312e+04          True
51     2.916630e+04          True
55     1.240492e+05          True
56     2.235539e+05          True
60     3.039042e+03          True
64     1.518834e+05          True
65     3.761167e+04          True
69     3.554834e+03          True
73     1.559710e+05          True
```

```
Training rows: 2741
Rows to predict: 887
✅ Validation RMSE: 23,019,138
✅ Validation R²: 0.7334
✅ Final imputed dataset saved as Deploying_New_Livestock_Count_Imputed.csv
Sample of imputed rows:
    Livestock_Count  Imputed_Flag
46     3.280052e+06          True
47     3.127952e+04          True
51     2.440448e+04          True
55     1.088246e+06          True
56     6.365034e+05          True
60     2.819436e+03          True
64     3.181021e+05          True
65     3.119307e+05          True
69     3.394004e+03          True
73     2.997279e+05          True
```

```
Training rows: 2741
Rows to predict: 887
✅ Validation RMSE: 30,391,704
✅ Validation R²: 0.5353
✅ Final imputed dataset saved as Deploying_New_Livestock_Count_Imputed_top_features.csv
Sample of imputed rows:
    Livestock_Count  Imputed_Flag
46     8.948107e+05          True
47     8.276688e+04          True
51     1.735755e+05          True
55     2.351756e+06          True
56     2.712882e+06          True
60     5.033948e+03          True
64     1.665968e+06          True
65     1.829586e+05          True
69     2.957691e+03          True
73     1.855344e+06          True
```

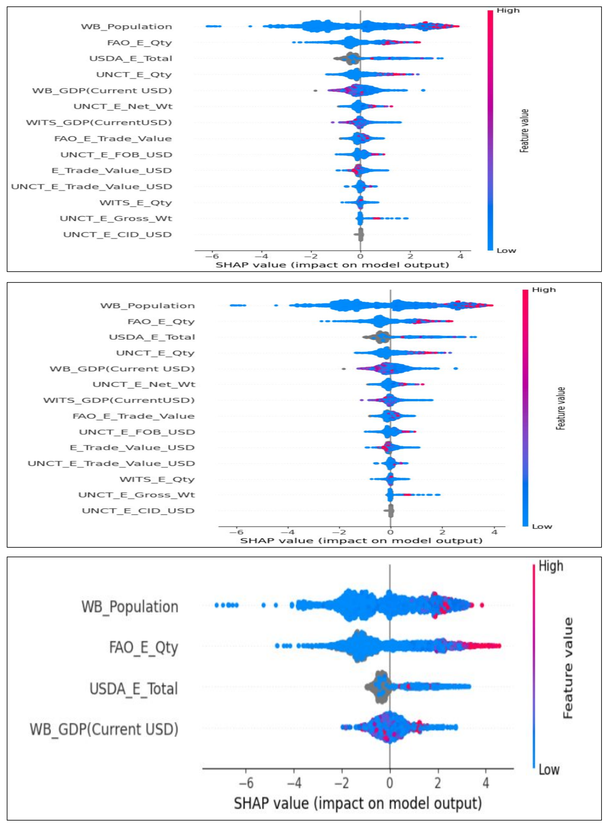


Figure 3: *Model evaluation and performance comparison of different imputation approaches for livestock count prediction, including full feature set, selected features, and top features, with validation RMSE and R² metrics reported for each approach: full features (RMSE: 22,791,185, R²: 0.7387), selected features (RMSE: 23,019,138, R²: 0.7334), top features (RMSE: 30,391,704, R²: 0.5353).*

Model Application and Real-World Testing

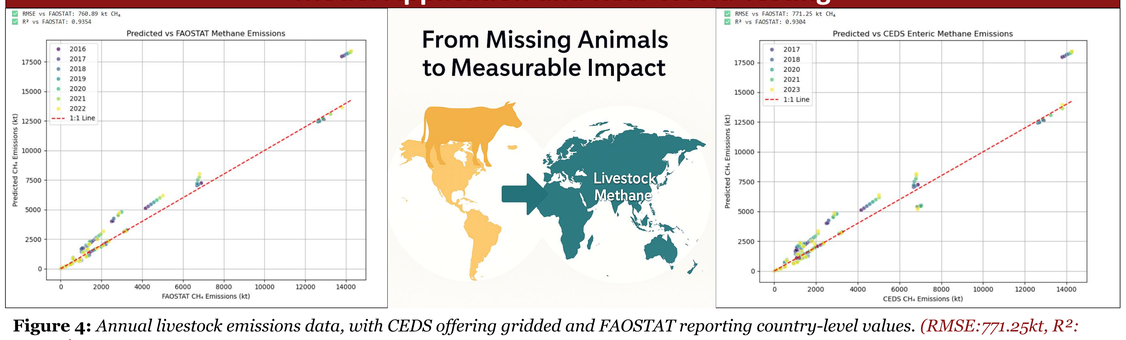


Figure 4: *Annual livestock emissions data, with CEDS offering gridded and FAOSTAT reporting country-level values. (RMSE:771.25kt, R²: 0.9304)*

Results

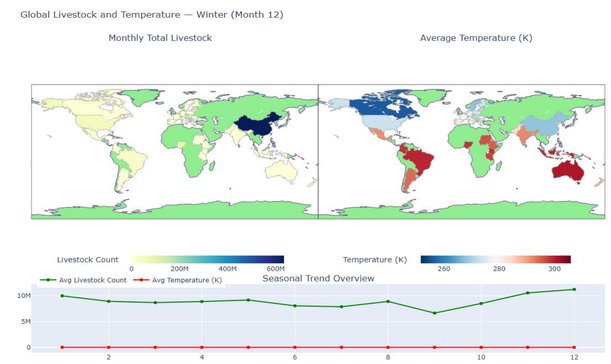


Figure 5: *Global spatial distribution of livestock and temperature in December (Winter). Monthly trends reveal seasonal cycles in livestock populations and temperature.*

Discussion

Our results show that data quality and preprocessing heavily influence livestock CH_4 inventory modeling. Log transformations improved model stability and reduced error magnitudes, but MAPE and approximate accuracy remain sensitive to extreme values. Incorporating trade and economic indicators enhanced predictions but also introduced variability due to global reporting inconsistencies.

Conclusions

An improved bottom-up inventory model for livestock methane emissions is feasible using multi-source activity data and machine learning for imputation. Early results demonstrate promising R^2 values but highlight challenges in prediction accuracy due to sparse and noisy datasets.

Future Directions

While the current system has yielded promising results, several challenges and opportunities remain:

- Low Temporal Resolution: Livestock data is mostly annual; higher frequency (monthly/weekly) would improve responsiveness.
- Coverage Gaps: Large data gaps in South America, Africa, Asia, and Oceania due to incomplete or inconsistent reporting.
- Data Enhancement: Add proxies (e.g., feed imports, production indices) and explore sub-national datasets to reduce aggregation errors.
- Model Improvement: Tune hyperparameters, test ensembles, and use metrics suited for skewed agricultural data.
- Integration: Generate monthly CH_4 estimates for top emitters and link to near-real-time monitoring frameworks.

Acknowledgements | References | Contact

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